

Rachit Sabharwal

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Professional Summary

Ph.D. Candidate (expected Aug 2026) and 6+ years building recommendation systems, behavioral and predictive models, and production ML pipelines across consumer goods, biotech, and population health. Co-authored a recommender-systems literature survey, shipped two end-to-end recsys applications (BioRec, PatRec), and modeled shopper and retailer behavior for a global CPG brand. Comfortable owning the full lifecycle from large-scale ETL through model validation, deployment, and stakeholder reporting.

Education

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| Ph.D. The University of Texas Health Science Center at Houston, Epidemiology and Biostatistics | Houston, TX
Expected Aug 2026 |
| <ul style="list-style-type: none">• Focus: Explainable AI• Certificates: Advanced Data Science | |
| Cert. The University of California, Berkeley, Software Development and Programming | Berkeley, CA
Apr 2025 |
| M.S. The University of Texas Health Science Center at Houston, Biostatistics | Houston, TX
May 2022 |
| B.S. University of Rochester, Environmental Health | Rochester, NY
May 2018 |

Experience

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| The University of Texas Health Science Center at Houston
Research Data Scientist | Houston, TX
Feb 2020 – present |
| <ul style="list-style-type: none">• Dissertation on explainable multimodal deep learning for severe TBI outcome prediction — systematic review of ML/DL methods, time-series deep learning on ICU physiological signals benchmarked against classical-ML baselines, and a multimodal model with XAI layers.• Co-authored a recommender-systems literature survey (KAIS 2023) and shipped BioRec, a biomedical recsys for academic conferences and journals, deployed via Streamlit as an interactive user-facing tool.• Built an agentic LLM pipeline (Groq, DeepSeek R1 70B, ellmer) that migrated the r4epi epidemiology textbook from R Markdown to Quarto, now adopted as official course material at UTHealth and TCU.• Led a team evaluating LLM feasibility for free-text physician-note classification in DETECT-RPC, informing the project's NLP pipeline design.• Cut data-munging turnaround 24x by re-architecting the TexasCARES ETL; stood up CI/CD reporting that ingests new data weekly, runs quality checks, and publishes stakeholder outputs without manual intervention.• Shipped a public-facing statewide COVID-19 dashboard (Python, Tableau) ingesting case and antibody-survey data continuously.• Co-authored 4 peer-reviewed publications on SARS-CoV-2 vaccine non-response and pediatric COVID comorbidities using survival analysis and mixed models on population-based cohorts. | |
| The Dow Chemical Company
ML R&D Intern | Lake Jackson, TX
May 2025 – Aug 2025 |
| <ul style="list-style-type: none">• Shipped an R Shiny decision-support app automating 50% of enterprise LCA workflows, estimated at \$15M in annual cost savings across Dow's global operations.• Benchmarked Bayesian UQ methods (Gaussian processes, BNNs, MC dropout) against frequentist baselines using calibration metrics (PICP, ECE), informing organizational guidance for predictive-model validation. | |

Genentech

Informatics R&D Intern

San Francisco, CA
May 2021 – Jan 2022

- Reproduced a published multimodal GNN benchmark, then prototyped a lower-complexity MLP combining molecular embeddings (ECFP fingerprints) with drug diffusion profiles in PyTorch Lightning, matching the published baseline within 10% at substantially reduced model complexity.

Procter & Gamble

Advanced Analytics Co-Op

Cincinnati, OH
Jan 2021 – Apr 2021

- Delivered predictive and customer-segmentation models on shopper and retailer behavior for global Fabric Care brand teams, informing corporate strategy around DTC ecommerce.
- Raised cross-functional analyst throughput 1.5x by building GCP big-data ETL pipelines using Apache Airflow and BigQuery.
- Raised codebase test coverage from 0% to 70% via containerization, unit testing, and agile practices.

Bristol Myers Squibb

Data Engineering Intern

San Francisco, CA
June 2020 – Aug 2020

- Built PatRec, a full-stack patent-recommendations app chaining chemical-entity NER, NCBI federated search, and TF-IDF cosine-similarity ranking over an 80M-patent corpus for internal BMS scientists.
- Designed PostgreSQL and Neo4j database architectures supporting multiple internal R&D analytical workloads.

Selected Publications

Scholarly recommendation systems: a literature survey

June 2023

Zhang Z, Patra BG, Sabharwal R, et al.

10.1007/s10115-023-01901-x (Knowledge and Information Systems)

BioRec: A Biomedical Recommendation System for Academic Conferences and Journals Sabharwal R

Apr 2022

(UTHealth School of Public Health, M.S. Thesis)

Baseline characteristics of SARS-CoV-2 vaccine non-responders in a large population-based sample

May 2024

Yaseen A, DeSantis SM, Sabharwal R, et al.

10.1371/journal.pone.0303420 (PLoS One)

Skills

Machine Learning & AI: Recommender Systems, Personalization, Customer Segmentation, Gradient Boosted Trees, Deep Learning, Embeddings, Multimodal Models, Explainable AI (SHAP), Time-Series Modeling, NLP, LLM Agentic Pipelines, GenAI APIs (Groq, Claude, Gemini, OpenAI-compatible), Bayesian Inference & UQ, Survival Analysis, Mixed Models

ML/AI Libraries: PyTorch, PyTorch Lightning, TensorFlow, Hugging Face, scikit-learn, Optuna, SHAP, tidymodels

Data & Cloud at Scale: Python, SQL, R, Bash; BigQuery, Apache Airflow, Spark/PySpark (familiar), Docker, Kubernetes, GitHub/GitLab CI/CD; GCP, AWS, Azure; PostgreSQL, pgvector, Neo4j

Domain: Consumer & Retail Analytics (CPG, DTC ecommerce), Real-World Health Data (EHR, claims, surveys), Clinical Outcomes Research, HIPAA-aware data handling